

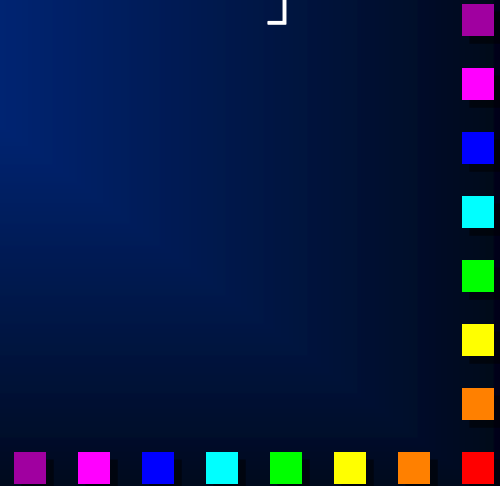
The CALL Verb

CALL Syntax

CALL $\left\{ \begin{array}{l} \text{ProgNameIdentifier} \\ \text{ProgNameLiteral} \end{array} \right\}$

$\left[\begin{array}{l} \text{USING} \left\{ \begin{array}{l} \left[\text{BY } \underline{\text{REFERENCE}} \right] \\ \text{BY } \underline{\text{CONTENT}} \end{array} \right\} \text{ParamIdentifier} \dots \end{array} \right\} \dots \right]$

$\left[\begin{array}{l} \left\{ \begin{array}{l} \text{ON } \left\{ \begin{array}{l} \underline{\text{EXCEPTION}} \\ \underline{\text{OVERFLOW}} \end{array} \right\} \text{StatementBlock} \\ \underline{\text{NOT ON }} \underline{\text{EXCEPTION}} \text{StatementBlock} \end{array} \right\} \left[\underline{\text{END - CALL}} \right] \end{array} \right]$



CALL Example.

CALL "DateValidate"
USING BY CONTENT TempDate
USING BY REFERENCE DateCheckResult.

IDENTIFICATION DIVISION.

PROGRAM-ID DateValidate IS INITIAL.

DATA DIVISION.

WORKING-STORAGE SECTION.

??????????????

LINKAGE SECTION.

01 DateParam PIC X(8).

01 DateResult PIC 9.

PROCEDURE DIVISION USING DateParam, DateResult.

Begin.

??????????????

??????????????

EXIT PROGRAM.

??????.

??????????????

CALL Parameters

CALL "ProgramName" USING P1, P2, P3, P4.

PROCEDURE DIVISION USING P2, P4, P1, P3.

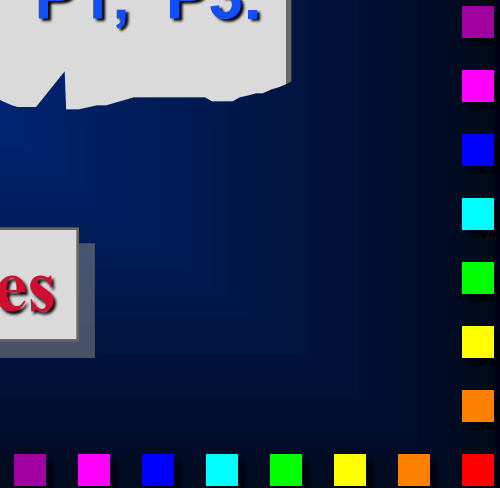


CALL Parameters

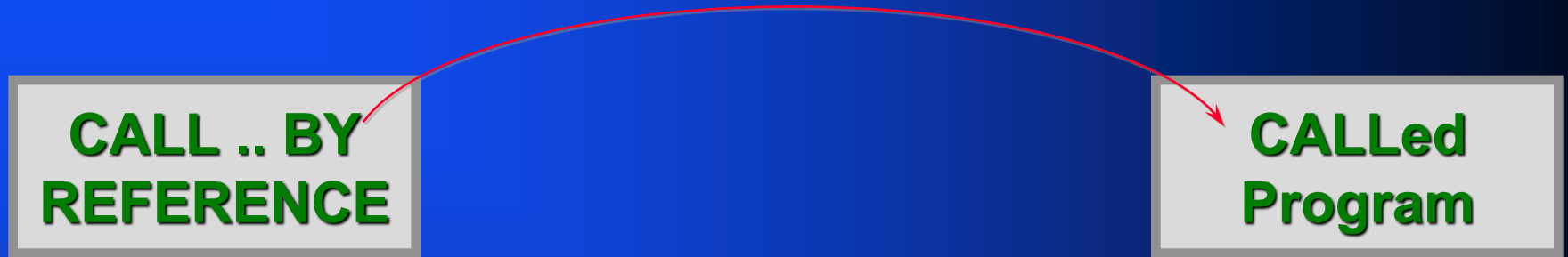
CALL "ProgramName" USING P1, P2, P3, P4.

PROCEDURE DIVISION USING P2, P4, P1, P3.

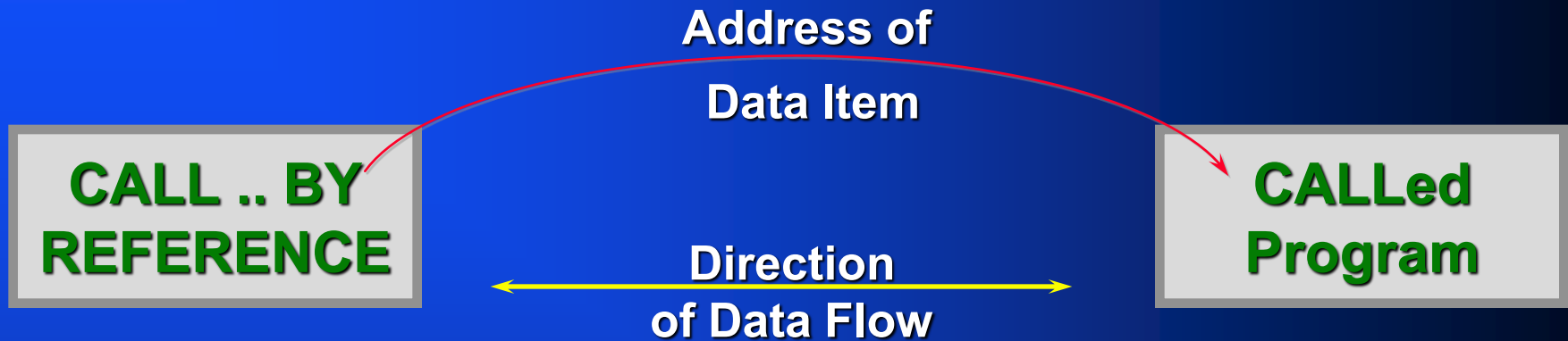
Positions Correspond - Not Names



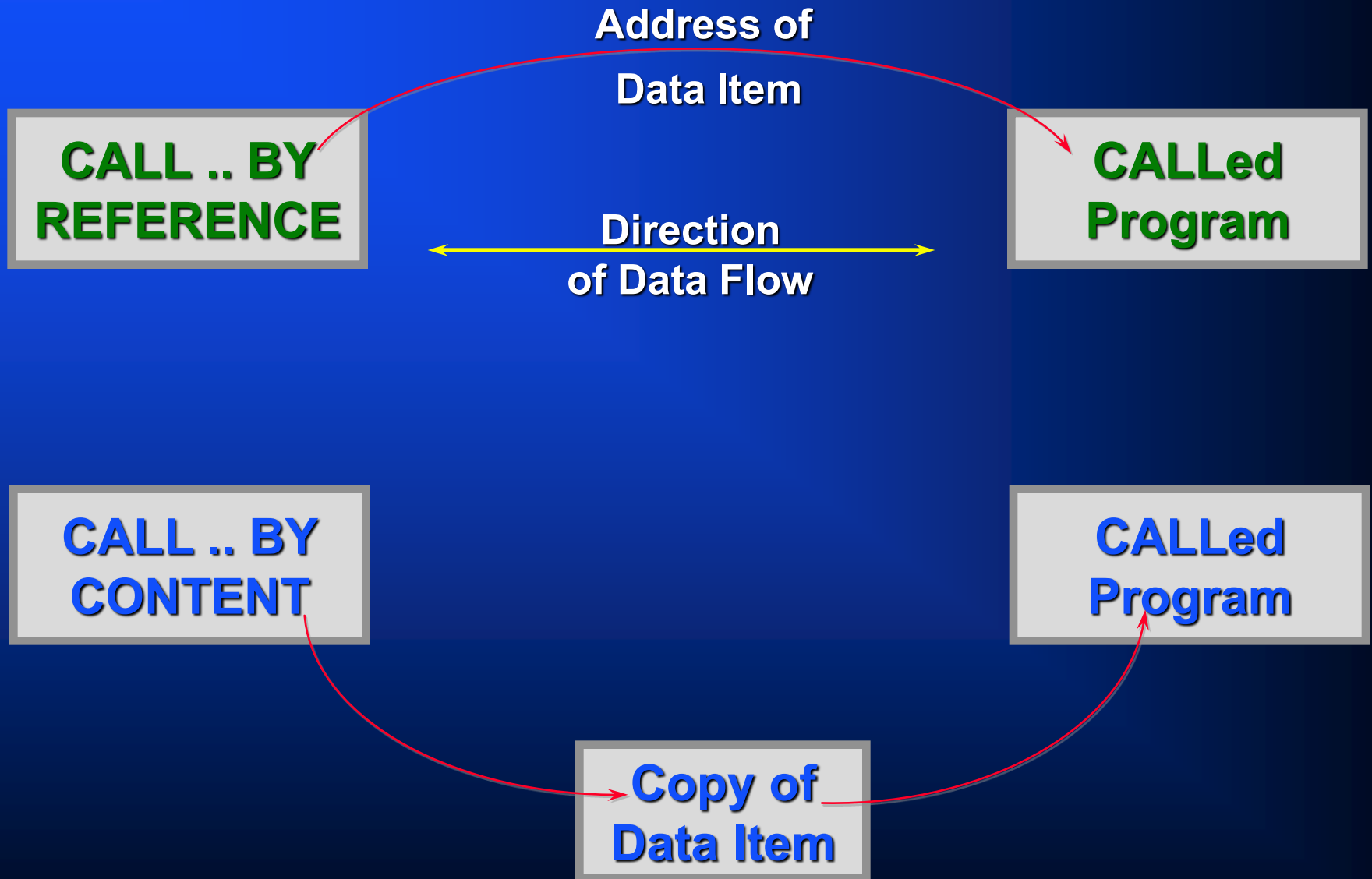
Parameter Passing Mechanisms



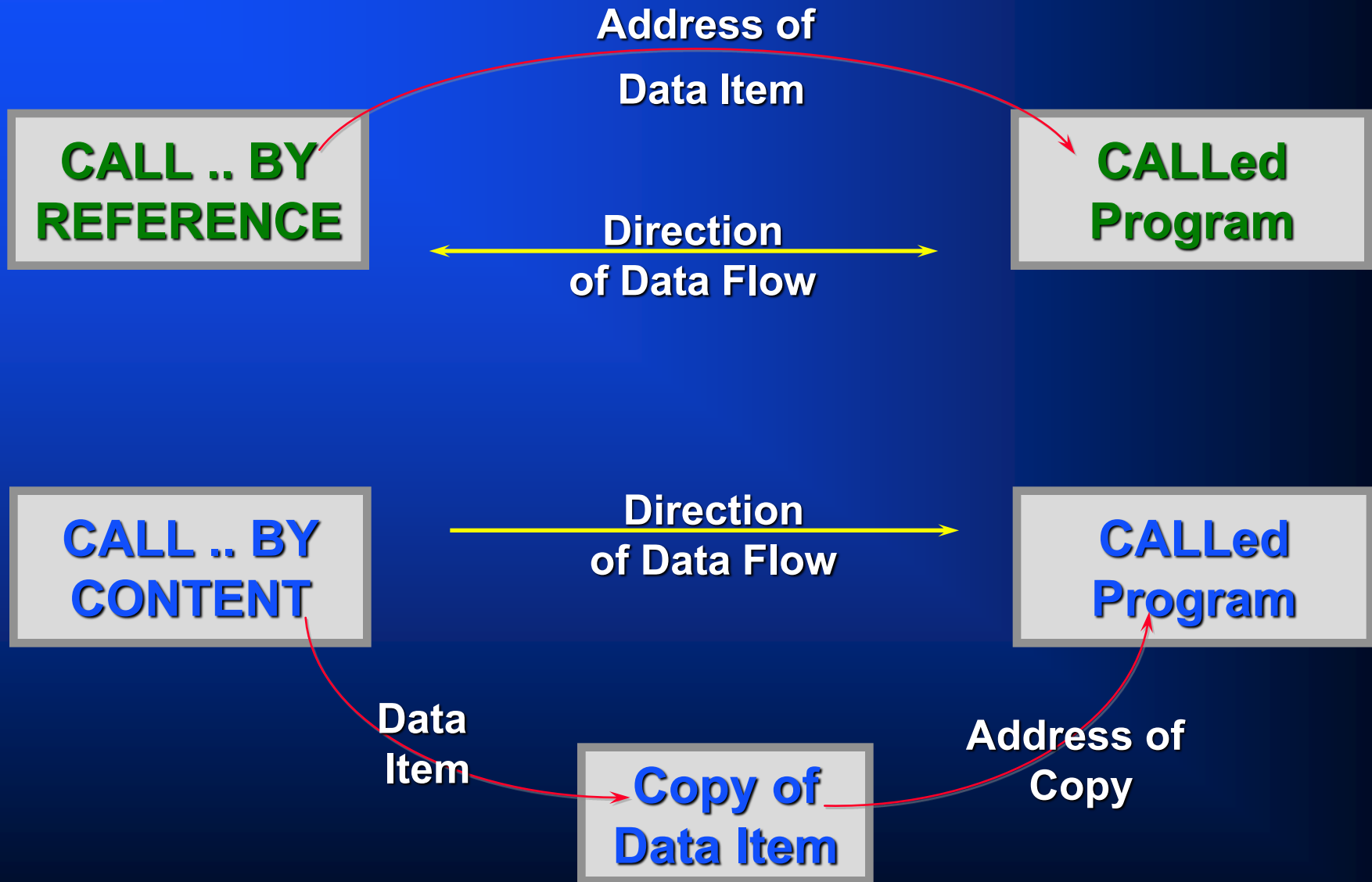
Parameter Passing Mechanisms



Parameter Passing Mechanisms



Parameter Passing Mechanisms



Avoiding "State Memory" - The IS INITIAL phrase.

```
    $ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Steadfast IS INITIAL.
```

12

Total = 62

```
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 RunningTotal    PIC 9(7) VALUE 50.
```

5

Total = 55

```
LINKAGE SECTION.  
01 ParamValue      PIC 99.
```

```
PROCEDURE DIVISION USING ParamValue.  
Begin.
```

12

Total = 62

```
    ADD ParamValue TO RunningTotal.  
    DISPLAY "Total = ", RunningTotal.  
EXIT PROGRAM.
```

Avoiding "State Memory" - The IS INITIAL phrase.

```
      $ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Fickle.  
  
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 RunningTotal      PIC 9(7) VALUE 50.  
  
LINKAGE SECTION.  
01 ParamValue        PIC 99.  
  
PROCEDURE DIVISION USING ParamValue.  
Begin.  
    ADD ParamValue TO RunningTotal.  
    DISPLAY "Total = ", RunningTotal.  
    EXIT PROGRAM.
```

12

Total = 62

5

Total = 67

12

Total = 79

The CANCEL command.

```
CALL "Fickle" USING BY CONTENT IncValue.  
CANCEL "Fickle"  
CALL "Fickle" USING BY CONTENT IncValue.
```

```
      $ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. Fickle.  
  
DATA DIVISION.  
WORKING-STORAGE SECTION.  
01 RunningTotal    PIC 9(7) VALUE 50.  
  
LINKAGE SECTION.  
01 ParamValue      PIC 99.  
  
PROCEDURE DIVISION USING ParamValue.  
Begin.  
    ADD ParamValue TO RunningTotal.  
    DISPLAY "Total = ", RunningTotal.  
    EXIT PROGRAM.
```

12

Total = 62

12

Total = 62

Contained Sub-Programs

```
      $ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. MainProgram.
      ? ? ? ? ? ? ? ? ?
01 TableItem IS GLOBAL.

PROCEDURE DIVISION.
      ? ? ? ? ? ? ? ? ?
      CALL PutToTable USING BY CONTENT DataItem
      ? ? ? ? ? ? ? ? ?
      CALL ReportFromTable.
      EXIT PROGRAM.

IDENTIFICATION DIVISION.
PROGRAM-ID. PutToTable.
      ? ? ? ? ? ? ? ? ?
END-PROGRAM PutToTable.

IDENTIFICATION DIVISION.
PROGRAM-ID. ReportFromTable.
      ? ? ? ? ? ? ? ? ?
END-PROGRAM ReportFromTable.
END-PROGRAM MainProgram.
```

Contained Sub-Programs

```
$ SET SOURCEFORMAT"FREE"
```

```
IDENTIFICATION DIVISION.
```

```
PROGRAM-ID. MainProgram.
```

```
    ? ? ? ? ? ? ? ? ?
```

```
01 TableItem IS GLOBAL.
```

```
PROCEDURE DIVISION.
```

```
    ? ? ? ? ? ? ? ? ?
```

```
    CALL PutToTable USING BY CONTENT DataItem
```

```
    ? ? ? ? ? ? ? ? ?
```

```
    CALL ReportFromTable.
```

```
    EXIT PROGRAM.
```

```
IDENTIFICATION DIVISION.
```

```
PROGRAM-ID. PutToTable.
```

```
    ? ? ? ? ? ? ? ? ?
```

```
END-PROGRAM PutToTable.
```

```
IDENTIFICATION DIVISION.
```

```
PROGRAM-ID. ReportFromTable.
```

```
    ? ? ? ? ? ? ? ? ?
```

```
END-PROGRAM ReportFromTable.
```

```
END-PROGRAM MainProgram.
```

Contained Sub-Programs

```
$ SET SOURCEFORMAT"FREE"
```

```
IDENTIFICATION DIVISION.
```

```
PROGRAM-ID. MainProgram.
```

```
  ? ? ? ? ? ? ? ? ?
```

```
01 TableItem IS GLOBAL.
```

```
PROCEDURE DIVISION.
```

```
  ? ? ? ? ? ? ? ? ?
```

```
  CALL PutToTable USING BY CONTENT DataItem
```

```
  ? ? ? ? ? ? ? ? ?
```

```
  CALL ReportFromTable.
```

```
  EXIT PROGRAM.
```

```
IDENTIFICATION DIVISION.
```

```
PROGRAM-ID. PutToTable.
```

```
  ? ? ? ? ? ? ? ? ?
```

```
END-PROGRAM PutToTable.
```

```
IDENTIFICATION DIVISION.
```

```
PROGRAM-ID. ReportFromTable.
```

```
  ? ? ? ? ? ? ? ? ?
```

```
END-PROGRAM ReportFromTable.
```

```
END-PROGRAM MainProgram.
```

The COMMON PROGRAM Phrase

```
      $ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. MainProgram.  
      ? ? ? ? ? ? ? ?  
01 TableItem IS GLOBAL.  
PROCEDURE DIVISION.  
      ? ? ? ? ? ? ? ?  
EXIT PROGRAM.
```

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. PutToTable.  
      ? ? ? ? ? ? ? ?  
      CALL ReportFromTable.  
      ? ? ? ? ? ? ? ?  
END-PROGRAM PutToTable.
```

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. ReportFromTable IS COMMON PROGRAM.  
      ? ? ? ? ? ? ? ?  
      CALL PutToTable USING ????  
      ? ? ? ? ? ? ? ?  
END-PROGRAM ReportFromTable.  
END-PROGRAM MainProgram.
```


The COMMON PROGRAM Phrase

```
      $ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. MainProgram.  
      ? ? ? ? ? ? ? ?  
01 TableItem IS GLOBAL.  
PROCEDURE DIVISION.  
      ? ? ? ? ? ? ? ?  
EXIT PROGRAM.
```

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. PutToTable.  
      ? ? ? ? ? ? ? ?  
      CALL ReportFromTable.  
      ? ? ? ? ? ? ? ?  
END-PROGRAM PutToTable.
```

YES



```
IDENTIFICATION DIVISION.  
PROGRAM-ID. ReportFromTable IS COMMON PROGRAM.  
      ? ? ? ? ? ? ? ?  
      CALL PutToTable USING ????  
      ? ? ? ? ? ? ? ?  
END-PROGRAM ReportFromTable.  
END-PROGRAM MainProgram.
```

The COMMON PROGRAM Phrase

```
$ SET SOURCEFORMAT"FREE"  
IDENTIFICATION DIVISION.  
PROGRAM-ID. MainProgram.  
    ? ? ? ? ? ? ? ?  
01 TableItem IS GLOBAL.  
PROCEDURE DIVISION.  
    ? ? ? ? ? ? ? ?  
EXIT PROGRAM.
```

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. PutToTable.  
    ? ? ? ? ? ? ? ?  
    CALL ReportFromTable.  
    ? ? ? ? ? ? ? ?  
END-PROGRAM PutToTable.
```

```
IDENTIFICATION DIVISION.  
PROGRAM-ID. ReportFromTable IS COMMON PROGRAM.  
    ? ? ? ? ? ? ? ?  
    CALL PutToTable USING ????  
    ? ? ? ? ? ? ? ?  
END-PROGRAM ReportFromTable.  
END-PROGRAM MainProgram.
```

```
graph TD
    NO[NO] --> EndMain[END-PROGRAM MainProgram.]
    YES[YES] --> CallPut[CALL PutToTable USING ???]
```

Creating Abstract Data Types?

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Stack.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 StackHolder IS GLOBAL.
   02 StackItem OCCURS 20 TIMES PIC X(10) .

PROCEDURE DIVISION USING ????.
Begin.
    EVALUATE TRUE
        WHEN PushStack CALL "Push" USING ???
        WHEN PopStack CALL "Pop" USING ???
    END-EVALUATE.
EXIT PROGRAM.

IDENTIFICATION DIVISION.
PROGRAM-ID. Push.
    ? ? ? ? ? ? ? ? ? ?
END-PROGRAM Push.

IDENTIFICATION DIVISION.
PROGRAM-ID. Pop.
    ? ? ? ? ? ? ? ? ? ?
END-PROGRAM Pop.
END-PROGRAM Stack.
```

Main

Stack

Push

Pop

Creating Abstract Data Types?

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Stack.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 StackHolder IS GLOBAL.
   02 StackItem OCCURS 20 TIMES PIC X(10) .
PROCEDURE DIVISION USING ????.
Begin.
    EVALUATE TRUE
        WHEN PushStack CALL "Push" USING ???
        WHEN PopStack CALL "Pop" USING ???
    END-EVALUATE.
EXIT PROGRAM.

IDENTIFICATION DIVISION.
PROGRAM-ID. Push.
    ? ? ? ? ? ? ? ? ? ?
END-PROGRAM Push.

IDENTIFICATION DIVISION.
PROGRAM-ID. Pop.
    ? ? ? ? ? ? ? ? ? ?
END-PROGRAM Pop.
END-PROGRAM Stack.
```

Main

Stack

Push

Pop

Creating Abstract Data Types?

```
$ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID. Stack.
DATA DIVISION.
WORKING-STORAGE SECTION.
01 StackHolder IS GLOBAL.
   02 StackItem OCCURS 20 TIMES PIC X(10).
PROCEDURE DIVISION USING ????.
Begin.
    EVALUATE TRUE
        WHEN PushStack CALL "Push" USING ???
        WHEN PopStack CALL "Pop" USING ???
    END-EVALUATE.
EXIT PROGRAM.

IDENTIFICATION DIVISION.
PROGRAM-ID. Push.
    ? ? ? ? ? ? ? ? ? ?
END-PROGRAM Push.

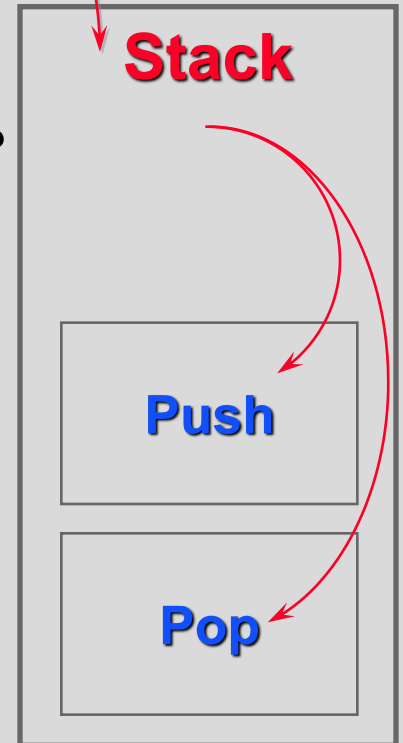
IDENTIFICATION DIVISION.
PROGRAM-ID. Pop.
    ? ? ? ? ? ? ? ? ? ?
END-PROGRAM Pop.
END-PROGRAM Stack.
```

Main

Stack

Push

Pop



The IS EXTERNAL phrase.

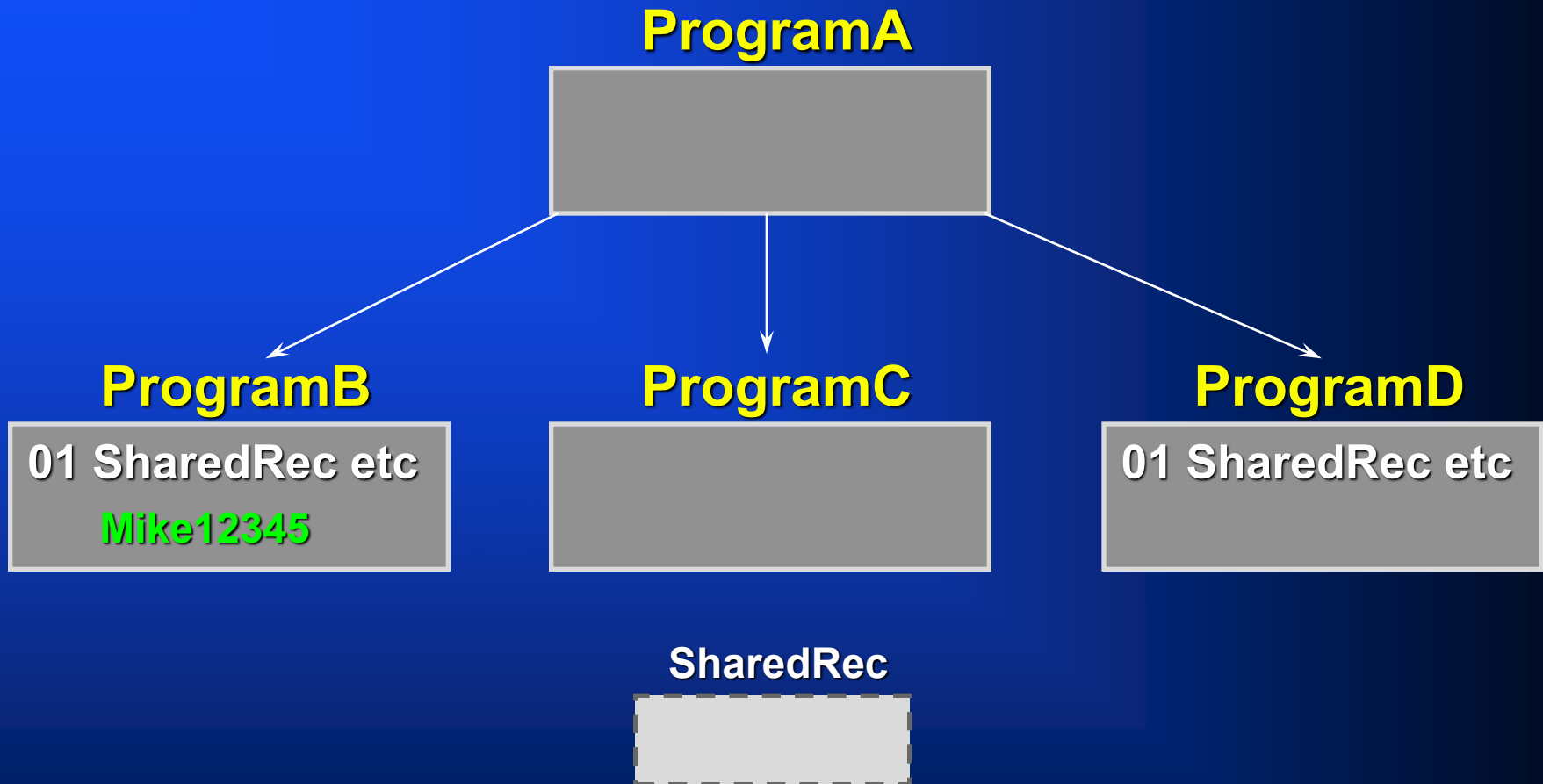
$\left\{ \begin{array}{l} \text{FDEntry} \\ \text{WSRecordEntry} \end{array} \right\} \text{ IS EXTERNAL}$

FD CommonFileArea IS EXTERNAL.

WORKING-STORAGE SECTION.
01 SharedRec IS EXTERNAL.
 02 PartA PIC X(4) .
 02 PartB PIC 9(5) .

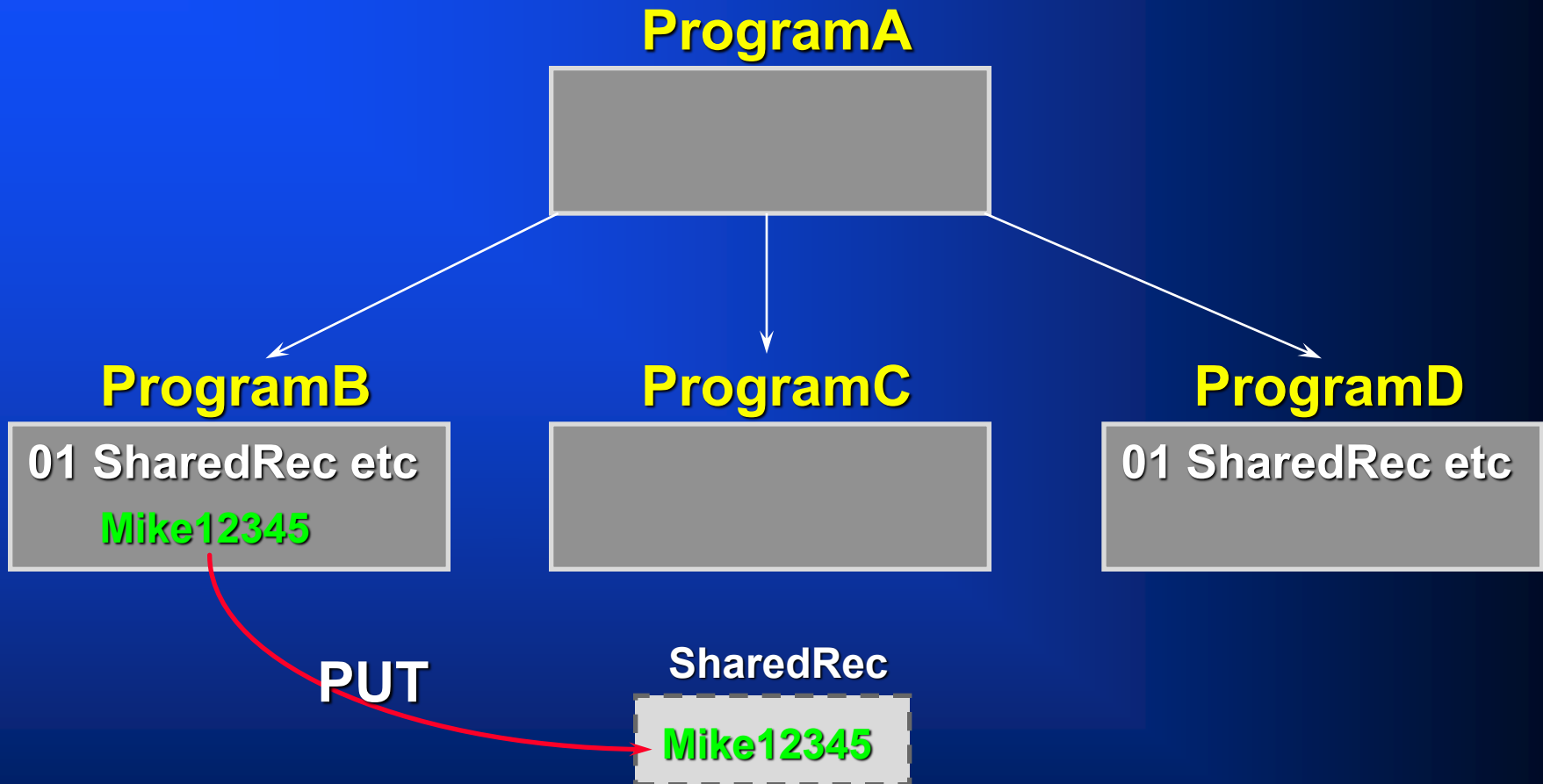


The IS EXTERNAL phrase.



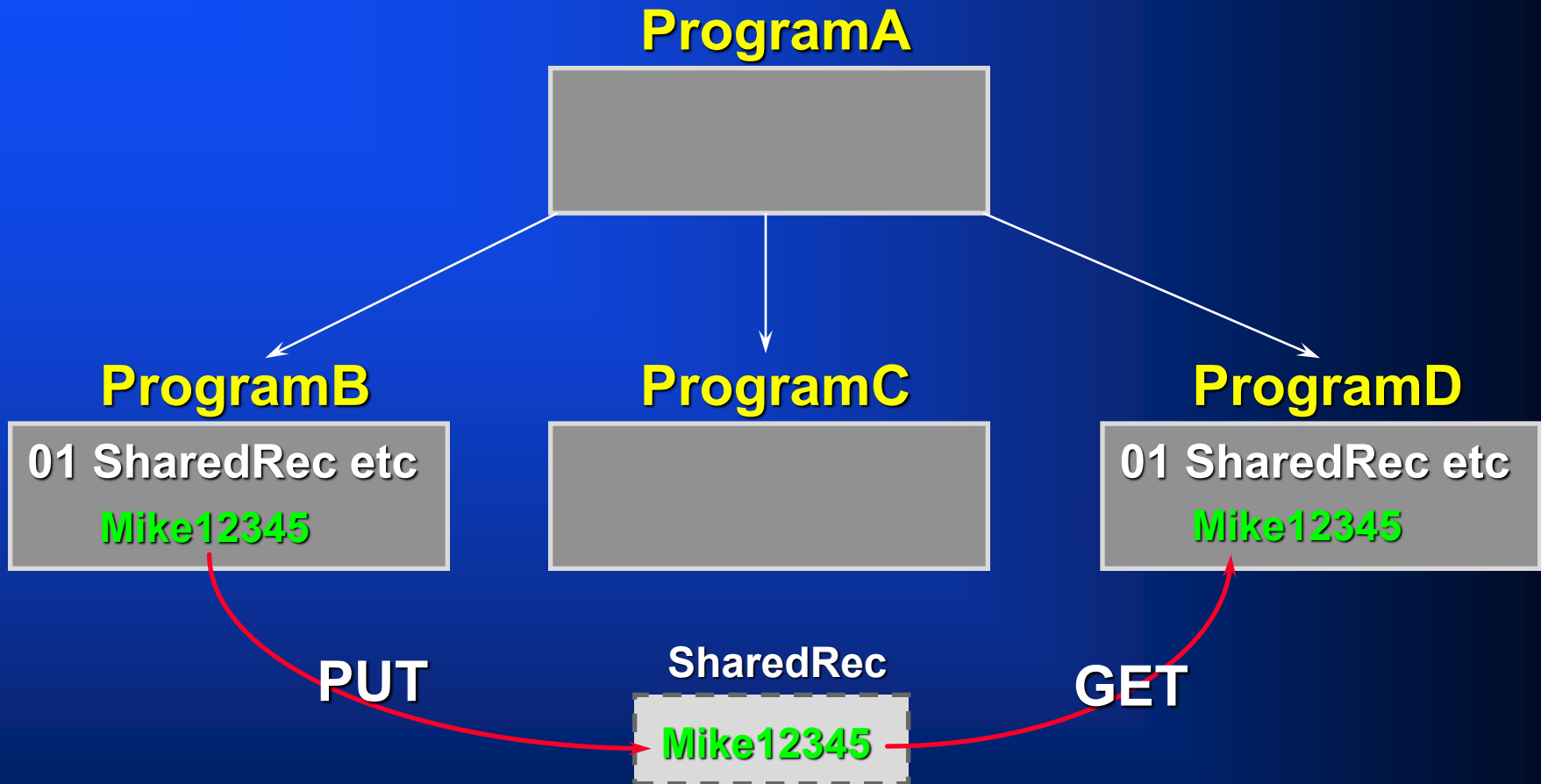
```
WORKING-STORAGE SECTION.  
01 SharedRec IS EXTERNAL.  
   02 PartA    PIC X(4) .  
   02 PartB    PIC 9(5) .
```

The IS EXTERNAL phrase.



```
WORKING-STORAGE SECTION.  
01 SharedRec IS EXTERNAL.  
   02 PartA    PIC X(4) .  
   02 PartB    PIC 9(5) .
```


The IS EXTERNAL phrase.



```
WORKING-STORAGE SECTION.  
01 SharedRec IS EXTERNAL.  
   02 PartA    PIC X(4) .  
   02 PartB    PIC 9(5) .
```